Multigrid Finite State Projection for the Reaction-Diffusion Master Equation

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The Chemical Master Equation: exact stochastic distributions

A well-mixed system: integer counts $\mathbf{n} \in \mathbb{Z}^S_{\geq 0}$ evolving by stochastic reactions.

The probability $p(\mathbf{n}, t)$ satisfies the **CME**:

$$\dot{p} = A\, p,$$

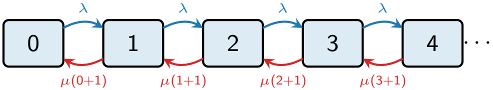
where A encodes every reaction event as a transition rate.

SSA vs. CME

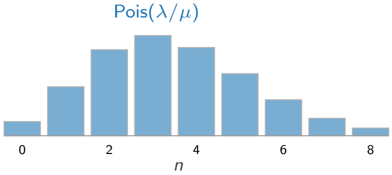
SSA samples individual trajectories.

CME gives the *exact* joint distribution $p(\mathbf{n}, t)$ directly — rare events, tails, and multimodal structure without ensemble noise.

Birth-death: $\emptyset \xrightarrow{\lambda} X \xrightarrow{\mu n} \emptyset$



Stationary distribution: Poisson(λ/μ).



FSP: truncate A to a finite support $\mathcal{S}$ and solve $\dot{p} = Ap$

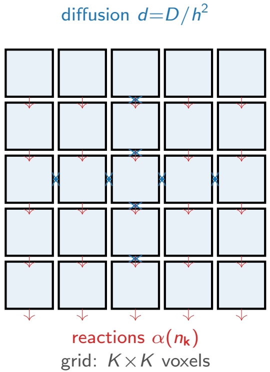
The RDME: reactions and diffusion on a spatial grid

Divide a 2-D domain into $K \times K$ voxels. Each voxel $\mathbf{k}$ holds a molecule count $n_{\mathbf{k}} \in \mathbb{Z}_{\geq 0}$.

Two types of events:

- **Reactions** inside voxel $\mathbf{k}$: propensity $\alpha(n_{\mathbf{k}})$.
- **Diffusion** between neighbors $\mathbf{k} \leftrightarrow \mathbf{l}$: rate $d = D/h^2$, moves one molecule.

The joint probability $p(\{n_{\mathbf{k}}\}, t)$ satisfies $\dot{p} = Ap$ **(RDME)**.



The curse of dimensionality

Joint state space: $\sim n_{\max}^{K^2}$ entries.
A 200×200 grid is **completely intractable** directly.

Key locality: diffusion only couples *adjacent* voxels.
This structure enables a principled decomposition.

Per-voxel CME with mean-field diffusion coupling

Key idea: diffusion couples voxels through their means.

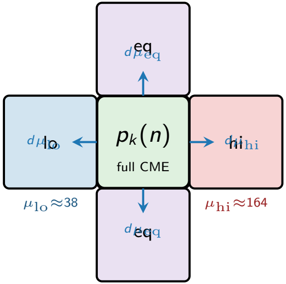
A molecule hops from voxel ℓ to voxel k at rate $d \cdot n_\ell$. Replace n_ℓ by its mean $\mu_\ell = \langle n_\ell \rangle$:

Mean-field RDME decomposition

$$\text{Diffusion in voxel } k: \quad \begin{array}{l} \text{birth at rate } d \sum_{\ell \sim k} \mu_\ell \\ \text{death at rate } d \cdot |\partial k| \cdot n_k \end{array}$$

Each voxel k evolves under an **effective 1-D CME**:

$$\dot{p}_k = Q_k(\{\mu_\ell\}_{\ell \sim k})\, p_k$$



mean-field from neighbours

Why it works: at the wavefront, equil voxels are sharp (delta-like at their basin), so μ_ℓ fully characterises them. The ADI pair step handles active-active coupling

Adaptive resolution: active vs. equilibrated voxels

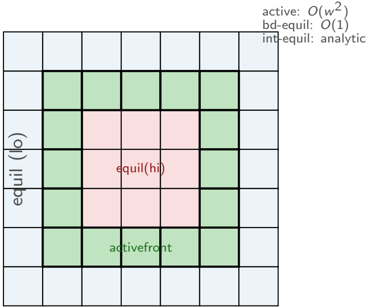
Active voxels (wavefront)

Full per-voxel CME via $\mathrm{expv}(dt, Q_k, p_k)$.
Tridiagonal generator Q_k — $O(w)$ Krylov cost where $w \approx 20$–50 is the active support width.

Equilibrated (equil) voxels

Distribution locked near a stable basin.
Boundary equil (adjacent to active): expv at reduced Krylov dimension $m_{\mathrm{eq}}=8$.
Interior equil: algebraic update only — $\mu_k \leftarrow (k_b + \mathrm{in})/(k_d + \mathrm{out})$, $p_k \leftarrow \mathrm{Pois}(\mu_k)$.
No matrix exponential needed.

ADI pair step (active-active pairs)



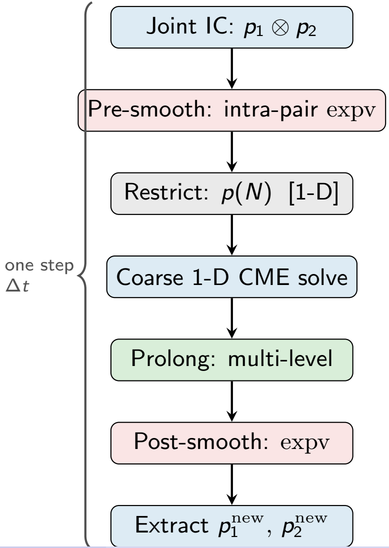
At peak: BD 200×200 grid — ∼5000 active voxels (thin ring), ∼35000 equil (interior), ∼0 equil (exterior, not yet reached).

Interior equil updated analytically: $O(1)$ per voxel.

The ADI pair V-cycle: exact 2-voxel RDME at the front

For each adjacent active pair (v_1, v_2):

1. Build sparse joint IC: $p_1 \otimes p_2$ (product state)
2. **Pre-smooth**: evolve intra-pair diffusion $v_1 \leftrightarrow v_2$ for $\tau_{\mathrm{pre}} = 0.1\,\Delta t$
3. **Restrict**: marginalise to total count $p(N) = \sum_{n_1+n_2=N} p(n_1, n_2)$ [1-D vector]
4. **Coarse solve**: 1-D CME on N with external boundary flux, for full Δt
5. **Prolong**: multi-level reconstruction of joint $p_{\mathrm{new}}(n_1, n_2)$ from $p_{\mathrm{post}}(N)$
6. **Post-smooth**: intra-pair diffusion for τ_{post}
7. Extract marginals p_1^{new}, p_2^{new}



ADI alternation

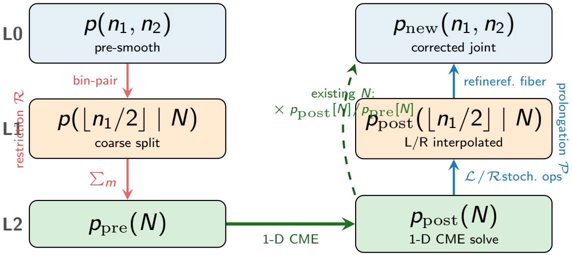
Multi-level prolongation: three-level hierarchy

The restriction erases the split.

Marginalising $p(n_1, n_2)$ to $p(N)$ loses all information about *how* N *molecules are distributed across* v_1 *and* v_2.

Three-level hierarchy on the split axis:

- **L0 (fine)**: $p(n_1 \mid N)$ exact conditional
- **L1 (mid)**: $p(\lfloor n_1/2 \rfloor \mid N)$ — pair consecutive n_1 into bins of width 2
- **L2 (coarse)**: $p(N)$ total count only



Prolongation rule

Existing N (in support of p_{pre}):

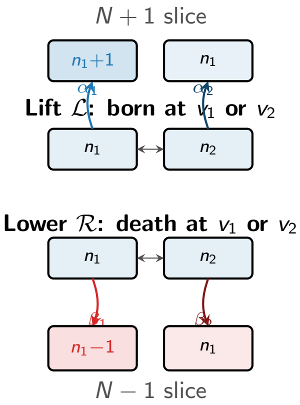
exact:

$$p_{\text{new}}(n_1, n_2) = p_{\text{pre}}(n_1, n_2) \times \frac{p_{\text{post}}[N]}{p_{\text{pre}}[N]}$$

New N (coarse solve added probability):

Why L/R and not product-convolution?

Product-convolution uses marginals $m_1(n_1) \cdot m_2(N - n_1)$ — a flat prior that ignores which reaction changed N.



Stochastic lifting $\mathcal{L}$: $N \to N+1$

When a molecule is born, it goes to v_1 or v_2 with probabilities determined by the actual propensities:

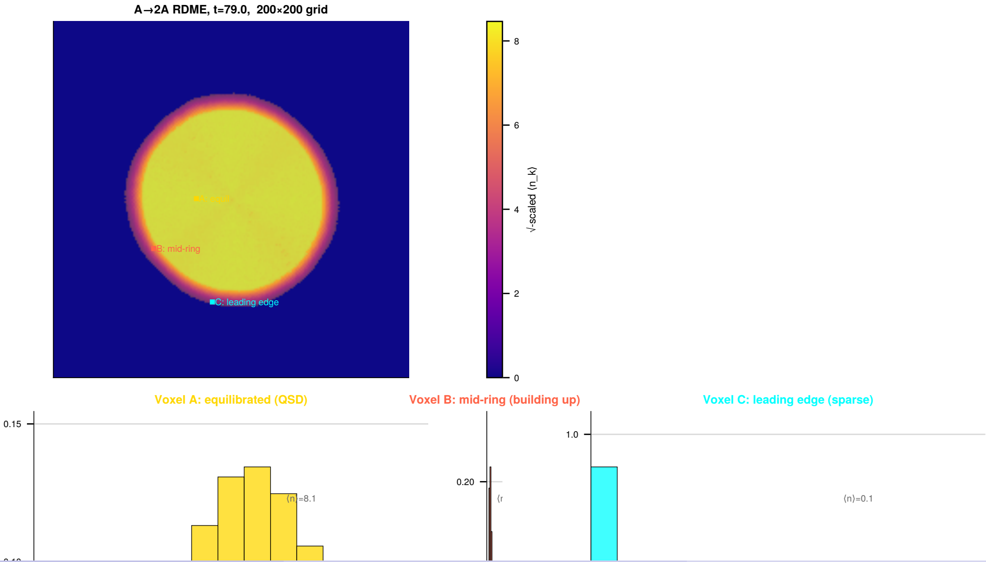
$$\alpha_1(n_1) = \frac{b_1(n_1)}{b_1(n_1) + b_2(n_2)}, \quad b_i(n) = \alpha_{\mathrm{rxn}}(n) + d\,\mu_i^{\mathrm{ext}}$$

$$\mathcal{L}[p(\cdot \mid N)](n_1) = \alpha_1(n_1 - 1)\, p(n_1 - 1 \mid N) + \alpha_2(n_1)\, p(n_1 \mid N)$$

Stochastic lowering $\mathcal{R}$: $N \to N-1$

Conditioned on a death event, molecule leaves from

Result: A→2A wavefront on a 200×200 RDME



Schlögl model:

$$2X \underset{c_2}{\overset{c_1}{\rightleftharpoons}} 3X, \qquad \emptyset \underset{c_4}{\overset{c_3}{\rightleftharpoons}} X$$

Two stable fixed points per voxel: $n_{\mathrm{lo}}\approx 38$, $n_{\mathrm{hi}}\approx 164$, unstable point $n_{\mathrm{un}}\approx 63$.

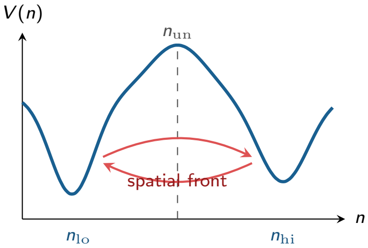
Why mean-field still works

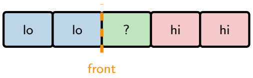
At the front, equil voxels are sharp (delta-like at n_{lo} or n_{hi}), so μ_ℓ fully describes them — mean-field is **exact** for equil-to-active coupling.

Active-active coupling handled by the ADI pair V-cycle exactly.

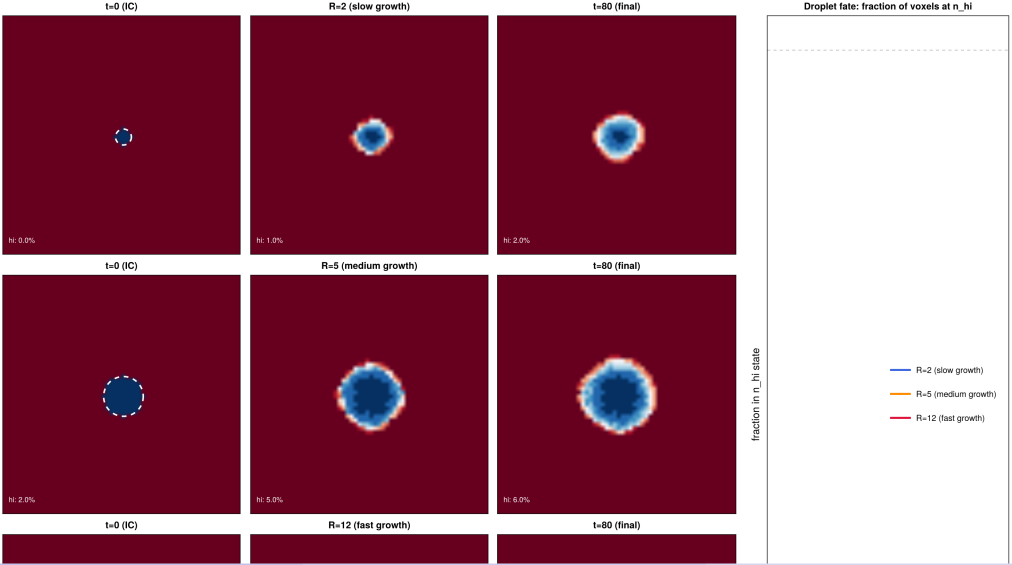
P(hi) colormap

Color each voxel by $P(n > n_{\mathrm{un}})$: 0 (equil-lo) $\to$





Result: Schlögl critical droplet on a 60×60 grid



Summary

1. Per-voxel CME with mean-field coupling.

The full RDME is never assembled. Diffusion is approximated by a Poisson in-flux at the neighbor mean — exact for sharp (equil) voxels. Each voxel evolves independently via expv on a compact tridiagonal generator.

2. ADI pair V-cycle: exact 2-voxel RDME at the front.

Adjacent active voxel pairs solved exactly (no mean-field between them). Row/column alternation (ADI) gives isotropic 2-D propagation. Pre/post-smooth equilibrate the intra-pair split; a 1-D coarse CME captures the total-count dynamics with external boundary flux.

3. Multi-level stochastic prolongation.

Three-level hierarchy on the split axis. Existing total counts N: exact multiplicative correction — no approximation. New N values: physics-derived stochastic $\mathcal{L}/\mathcal{R}$ operators propagate the conditional fiber from the nearest known N, then a within-bin refinement restores fine resolution. Strictly better than product-convolution for bistable and nonlinear systems.